



# *INDIA.RUNS*

Build what next India runs on

Team Name: EvidenceRank AI

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Problem Statement : Intelligent Candidate Discovery & Ranking Challenge

## Solution Overview

**EvidenceRank AI** is an evidence-driven candidate intelligence system that ranks candidates based on demonstrated experience, technical fit, behavioral signals, hireability, and profile credibility rather than simple keyword matching.

**The system combines:**

- Career-history evidence analysis
- Retrieval & ranking expertise detection
- Behavioral signal scoring
- Risk & honeypot detection
- Hidden-gem candidate discovery
- Explainable candidate ranking

**Output:** Top 100 highly relevant and hireable candidates for the given job description.

| Traditional ATS Systems        | EvidenceRank AI                           |
|--------------------------------|-------------------------------------------|
| Keyword matching               | Evidence-based matching                   |
| Skills section driven          | Career-history driven                     |
| Easy to game with buzzwords    | Contradiction & trap detection            |
| Ignores candidate availability | Uses behavioral hireability signals       |
| Misses hidden gems             | Recovers implicit domain expertise        |
| Black-box ranking              | Explainable reasoning for every candidate |

**Key Innovation:**

**We rank candidates based on what they have actually built and demonstrated, not merely on the keywords they claim.**

## JD Understanding & Candidate Evaluation

### Explicit Requirements

- Retrieval, Ranking & Matching Systems
- Embeddings & Semantic Search
- Vector Databases (FAISS, Milvus, OpenSearch, etc.)
- Python & Production Engineering
- Evaluation Metrics (NDCG, MAP, MRR)
- Production ML Deployment
- 5–9 Years Relevant Experience

### Implicit Requirements

- Product-first mindset over research-only experience
- Demonstrated system-building capability
- Real-world deployment experience
- Active job-market participation
- Strong recruiter engagement & availability
- Ability to ship and iterate quickly

EvidenceRank AI evaluates demonstrated career evidence, behavioral hireability, and profile credibility to identify the best candidates—not merely those with the highest keyword overlap.

### Technical Evidence Signals

- Career history and role progression
- Search, ranking, recommendation, and retrieval projects
- Production deployment evidence
- Vector search and embedding infrastructure experience
- Evaluation framework exposure

### Behavioral & Hireability Signals

- Open-to-work status
- Recent platform activity
- Recruiter response rate
- Interview completion rate
- Notice period and relocation preference
- GitHub activity and skill assessments

### Credibility & Risk Signals

- Profile completeness
- Verified contact information
- Contradiction detection
- Timeline consistency checks
- Honeypot and keyword-stuffing detection

## Ranking Methodology

### Candidate Retrieval

#### Broad Recall Layer

Keyword expansion & domain synonyms  
Semantic retrieval of related experience  
Hidden-gem candidate discovery  
High-recall candidate pool generation

**Output:** Top relevant candidate pool

### Evidence Extraction & Validation

#### Candidate Evidence Graph

Career history analysis  
Project & achievement extraction  
Skill verification  
Behavioral signal extraction

#### Evidence Priority:

Career History > Projects > Skills > Headline

### Multi-Dimensional Scoring

#### Technical Fit Score

- Retrieval & ranking experience
- Recommendation/search systems
- Embeddings & vector databases
- Production ML deployment
- Evaluation frameworks (NDCG, MAP, MRR)

#### Hireability Score

- Open-to-work status
- Recruiter response rate
- Platform activity
- Notice period
- Interview completion rate

#### Risk Score

- Keyword stuffing
- Timeline inconsistencies
- Title-skill mismatch
- Honeypot indicators
- Inactive profiles

### Final Ranking Formula

Final Score =  
(Technical Fit  
+ Retrieval & Ranking Proof  
+ Product/Production Fit  
+ Behavioral Hireability  
+ Location/Notice Fit)  
- Risk Penalties

Final Rank Score = Fit × (1 - Risk Score)

### Re-Ranking & Calibration

- Pairwise Top-500 comparison
- Hidden-gem boosting
- Diversity & quality calibration
- Top-100 optimization

EvidenceRank AI combines Technical Fit, Hireability, and Risk into a unified ranking framework, ensuring candidates are both capable and realistically hireable.

## Explainability & Data Validation

### Explainable Ranking Decisions

Every ranking decision is linked to verifiable evidence extracted from the candidate profile through the Candidate Evidence Graph.

#### Evidence Hierarchy

Career History Proof



Project / Achievement Evidence



Skills Validation



Headline & Summary Claims

**Result:** Every ranking decision can be traced back to specific candidate evidence.

### Hallucination Prevention & Reasoning Validation

EvidenceRank AI generates explanations only from validated profile attributes.

#### Validation Rules

- Uses profile-supported facts only
- References actual skills, experience, and behavioral signals
- Includes strengths and concerns when applicable
- Prevents unsupported assumptions
- Maintains rank-consistent reasoning

**Result:** Transparent, factual, and defensible candidate explanations.

### Data Quality & Suspicious Profile Detection

#### Risk Detection Engine

##### Detects and penalizes:

- Keyword stuffing
- Title–skill inconsistencies
- Impossible career timelines
- Behavioral anomalies
- Inactive candidates
- Honeytrap indicators

#### Candidate Trust Assessment

Technical Fit

+

Behavioral Credibility

+

Profile Consistency

=

Trustworthy Ranking

EvidenceRank AI does not merely rank candidates—it validates evidence, detects inconsistencies, and produces explainable rankings that recruiters can trust.

**End-to-End Workflow**

Job Description



Hiring Profile



Semantic Candidate Retrieval



Evidence Graph



Multi-Dimensional Scoring



Re-Ranking & Calibration



Top-100 Ranked Candidates

EvidenceRank AI transforms the job description into a structured hiring profile and retrieves a broad candidate pool using semantic matching and domain-aware search.

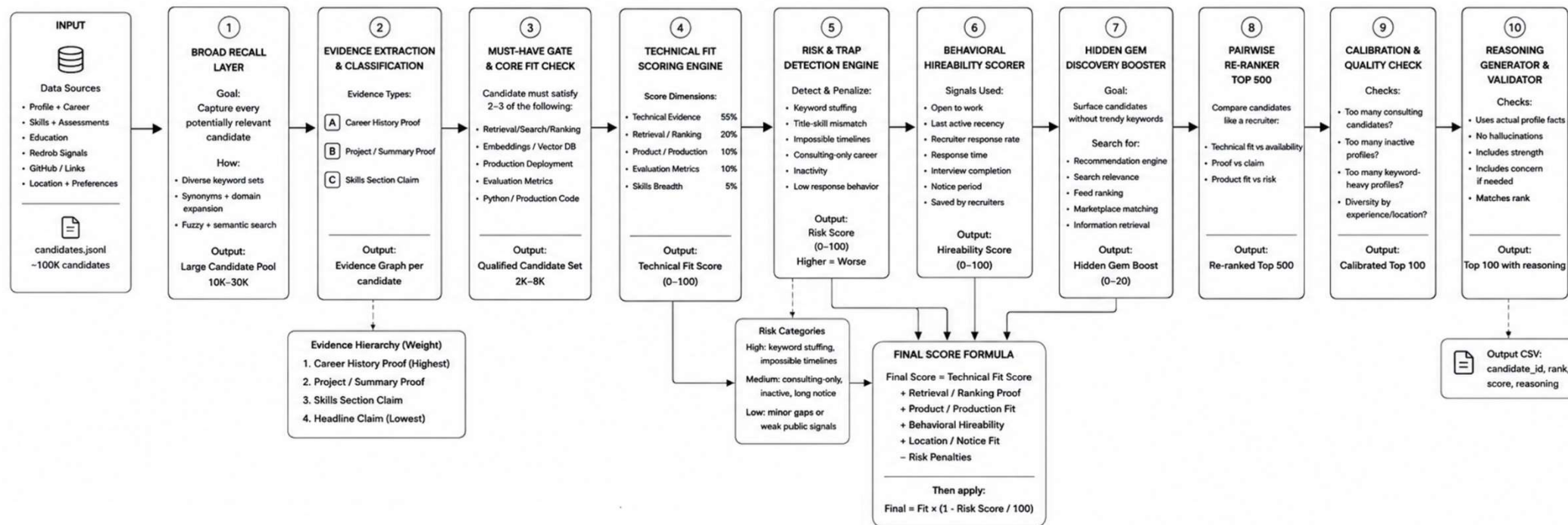
Each candidate is evaluated through an evidence graph that validates career achievements, technical expertise, behavioral signals, and profile credibility.

The system then combines Technical Fit, Hireability, and Risk scores, followed by calibration and re-ranking, to generate an explainable Top-100 candidate ranking aligned with the hiring intent rather than simple keyword overlap.

## System Architecture

## Redrob Candidate Ranking Pipeline (v5)

From 100K Raw Candidates to Top 100 Highly Reliable Matches



## Results & Performance

- What results or insights demonstrate ranking quality?

- Processed 100,000 candidate profiles and generated a ranked Top-100 shortlist.
- Multi-stage filtering combines technical fit, behavioral hireability, risk assessment, and hidden-gem discovery.
- Every shortlisted candidate includes human-readable reasoning for recruiter transparency.
- Evidence-based ranking reduces reliance on keyword matching and prioritizes verified candidate signals

- How does your solution meet the challenge's runtime and compute constraints?

- End-to-end pipeline completed in approximately **5 minutes** on Google Colab.
- Processed **100,000 candidates (~333 candidates/second)**.
- Memory-efficient Top-K ranking strategy avoids loading unnecessary intermediate results.
- Scalable design supports larger candidate pools through parallel processing and distributed deployment.

| Metric                     | Result              |
|----------------------------|---------------------|
| Candidates Processed       | 100,000             |
| Final Ranked Candidates    | 100                 |
| End to End Processing Time | ~5 min              |
| Throughput                 | ~333 candidates/sec |
| Candidate Reduction        | 99.9%               |
| Explainable Rankings       | 100%                |

**Note:** Runtime measured on Google Colab using the provided 100,000-candidate dataset. Reported time includes data ingestion, feature extraction, scoring, re-ranking, reasoning generation, and final CSV creation

Technologies Used

- What technologies, frameworks, and tools were used and why were they selected for this solution?

| Technology                   | Purpose                                                  |
|------------------------------|----------------------------------------------------------|
| Python                       | Core implementation of the ranking pipeline              |
| JSONL Processing             | Efficient handling of 100,000 candidate profiles         |
| Heap Queue (Top-K Selection) | Memory-efficient retrieval of top candidates             |
| Rule-Based AI Scoring Engine | Technical fit, hireability, risk, and hidden-gem scoring |
| Pandas                       | Data processing and CSV generation                       |
| Google Colab                 | Cloud-based execution environment                        |
| CSV Output Pipeline          | Submission-ready ranked candidate export                 |

Why These Technologies?

- Python provides rapid development and strong data-processing capabilities.
- Heap-based ranking enables scalable Top-K retrieval without storing all candidates in memory.
- Rule-based scoring ensures explainability and recruiter trust.
- Pandas + CSV provide simple integration with existing recruitment workflows.
- Google Colab offers accessible cloud compute for large-scale candidate processing.

Design Principle: Prioritized explainability, scalability, and recruiter trust over black-box ranking approaches.

## Submission Assets

### Source Code

EvidenceRank AI v5 Candidate Ranking Pipeline

End-to-end implementation of retrieval, scoring, risk filtering, re-ranking, and reasoning generation

### Output Deliverables

top\_100\_candidates.csv

Ranked candidate list with scores and explainable reasoning

### Documentation

Architecture Diagram

Workflow Diagram

AI Decision Flow

Technical Design Presentation

### Execution Environment

Google Colab implementation

Processed 100,000 candidate profiles in ~5 minutes

### Repository

GitHub: *(insert repository link if you upload it)*



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# THANK YOU